Hall Effect in n- and p-Type Germanium

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The Hall voltage and the resistance of n- and p-type germanium samples were measured against control current, magnetic flux density and, for the p-type sample, temperature, to determine the sign, density and mobility of the majority carriers. Opposite Hall-voltage signs identified electrons and holes. Three field sweeps per sample at 30 mA gave carrier densities of $(6.25 \pm 0.05) \times 10^{20}\mathrm{m}^{-3}$ (electrons) and $(8.60 \pm 0.03) \times 10^{20}\mathrm{m}^{-3}$ (holes) and Hall mobilities of (0.363 ± 0.002) and $(0.287 \pm 0.001)\mathrm{m}^2\mathrm{V}^{-1}\mathrm{s}^{-1}$. These uncertainties are statistical; the nominal sample dimensions were not measured. The p-type Hall voltage was sublinear in field, so its Hall coefficient depends on the fit range by up to 6%. The magnetoresistance mobility matched the Hall mobility for n-type, $(0.39 \pm 0.01)\mathrm{m}^2\mathrm{V}^{-1}\mathrm{s}^{-1}$, but was twice it for p-type, $(0.565 \pm 0.004)\mathrm{m}^2\mathrm{V}^{-1}\mathrm{s}^{-1}$. These p-type features, with a hole Hall mobility 50% above the lattice drift mobility, are consistent with two hole populations of different mobility. A zero-field offset proportional to current accounts for current-sweep slopes 12% (n-type) and 4% (p-type) above those expected at the set field. On heating, the p-type Hall voltage changed sign at 105°C, within the 91 to 117°C range estimated for intrinsic electrons to outweigh the measured hole density.

1. INTRODUCTION

The electrical behaviour of a semiconductor is set by its majority carriers: their sign, their density and their mobility, the drift velocity they acquire per unit electric field. A resistance measurement gives only the product of density and mobility. The Hall effect separates them. A magnetic field perpendicular to the current deflects the moving carriers sideways until a transverse electric field balances the magnetic force, and the transverse (Hall) voltage that results has the sign of the carriers and a size inversely proportional to their density [1,2].

This experiment followed the PC2193 manual [1] on two doped germanium samples, one n-type and one p-type. The Hall voltage was measured against control current and against magnetic field to obtain the carrier sign, the Hall coefficient, the carrier density and the Hall mobility. The rise of the sample resistance with field (magnetoresistance) gave a second estimate of mobility, which the manual expects to differ from the first. Finally, the Hall voltage of the p-type sample was followed during heating, through the temperature at which thermally generated electrons begin to dominate.

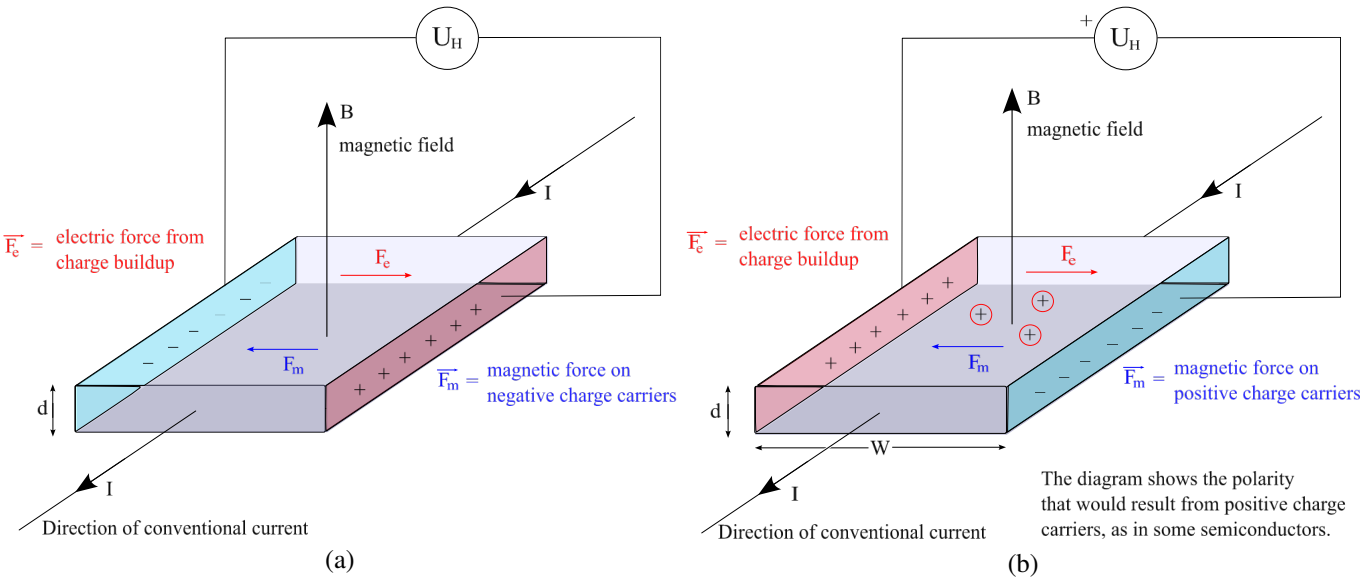


FIG. 1. Hall effect in a sample carrying a current in a magnetic field, for (a) negative and (b) positive charge carriers. The magnetic force F_m pushes the carriers of either sign towards the same side face, which charges with their sign; the resulting electric force F_e balances F_m in steady state. The Hall voltage across the sample therefore has opposite polarity in the two cases. Redrawn after HyperPhysics [3].

2. THEORY

2.1 Hall Coefficient, Carrier Density and Mobility

A carrier of charge q drifting with velocity $\boldsymbol{v}$ in a field $\boldsymbol{B}$ experiences the Lorentz force $\boldsymbol{F} = q\boldsymbol{v} \times \boldsymbol{B}$. Electrons and holes drift in opposite directions for the same current, so the force pushes both towards the same edge of the sample, and the sign of the charge that builds up there gives the sign of the carriers (Fig. 1). In steady state the transverse field cancels the magnetic force. For a sample of thickness d along B, the Hall voltage is then

$$U_H = H\frac{IB}{d}, \quad H = \pm\frac{1}{en}, \tag{1}$$

where H is the Hall coefficient ($\mathrm{m^3\,C^{-1}}$), e the elementary charge, n the carrier density, and the sign is negative for electrons and positive for holes [1,2]. Eq. (1) assumes one type of carrier with a velocity-independent scattering time; in general the right-hand side is multiplied by a Hall factor r_H of order one that depends on the scattering mechanism ($3\pi / 8 \approx 1.18$ for acoustic-phonon scattering) [4]. Like the manual, this report takes $r_H = 1$. The conductivity at zero field follows from the zero-field resistance R_0,

$$\sigma_0 = \frac{L}{R_0 w d}, \tag{2}$$

and the Hall mobility is $\mu_H = |H|\,\sigma_0$ [1]. In a field sweep at fixed current, $H = (\mathrm{d}U_H / \mathrm{d}B)d / I$ and $R_0 = U_{p,0} / I$, where $U_{p,0}$ is the zero-field sample voltage. Substituting gives

$$\mu_H = \left|\frac{\mathrm{d}U_H}{\mathrm{d}B}\right| \frac{L}{w U_{p,0}}, \tag{3}$$

which contains neither the thickness nor the current. The carrier density, $n = I / (ed\,|\,\mathrm{d}U_H / \mathrm{d}B|)$, depends on both.

2.2 Magnetoresistance

The manual describes the increase of resistance with field for a single type of carrier as

$$R = R_0\left[1 + (\mu B)^2\right], \tag{4}$$

and identifies μ with the carrier mobility [1]. This relation is an approximation. In the simplest model, one carrier with a single scattering time in a long sample, the Hall field cancels the magnetic force for every carrier and the resistance does not change with field [2]. A magnetoresistance appears when carriers have a spread of velocities or scattering times, when carriers of different mobility contribute, or when the current contacts at the ends short-circuit the Hall field, which matters for a sample only twice as long as it is wide [4]. The mobility μ_{MR} fitted with Eq. (4) is therefore an effective parameter, not necessarily equal to μ_H.

2.3 Two Carriers and the Sign Reversal

When electrons (density n, mobility μ_n) and holes (p, μ_p) both conduct, the Hall coefficient becomes

$$H = \frac{p\mu_p^2 - n\mu_n^2}{e\left(p\mu_p + n\mu_n\right)^2} \tag{5}$$

[4]. In a p-type sample at room temperature $p \gg n$ and $H > 0$. On heating, electron–hole pairs are generated across the band gap, with $np = n_i^2$, where the intrinsic density is

$$n_i = \sqrt{N_c N_v}\left(\frac{T}{300\ \mathrm{K}}\right)^{\frac{3}{2}} \exp\!\left(-\frac{E_g}{2k_B T}\right). \tag{6}$$

Here N_c and N_v are the effective densities of states at 300 K, k_B is Boltzmann's constant, and the band gap E_g falls with temperature [5]. With charge neutrality $p = n + N_A$, where N_A is the acceptor density, Eq. (5) vanishes when $p = b^2 n$, with $b = \mu_n / \mu_p$. This happens at the temperature where

$$n_i = N_A \frac{b}{b^2 - 1}. \tag{7}$$

Because electrons in Ge are more mobile than holes ($b > 1$), the Hall voltage of p-type Ge changes sign while holes are still the more numerous carrier.

3. METHODS

3.1 Apparatus

The apparatus (Fig. 2) is the PHYWE COBRA3 Hall effect set listed in the manual [1,6]. The n- and p-type samples are germanium plates on carrier boards, of nominal length $L = 20$ mm, width $w = 10$ mm and thickness $d = 1$ mm [6]; these dimensions were not measured. The plane pole pieces are 30 mm × 30 mm × 48 mm. The Hall effect module supplies a control current of up to ±30 mA, measures U_H across the sample

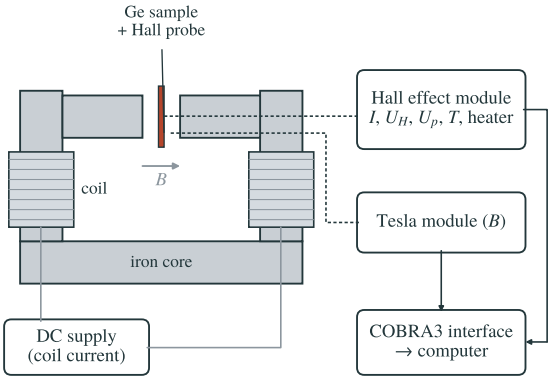


FIG. 2. Apparatus. The Ge sample on its carrier board sits in the Hall effect module, between the plane pole pieces of an electromagnet (U-shaped laminated iron core, two 600-turn coils driven by a DC supply). A tangential Hall probe in the same gap, read by the Tesla module, measures B at the sample. The module supplies the control current I and reads the Hall voltage U_H, the sample voltage U_p and the sample temperature T; the COBRA3 interface records them on a computer.

width and the sample voltage U_p along its length, and contains a heater and a temperature sensor. The field was set by adjusting the coil supply and read with the Hall probe through the Tesla module. The manual provides a knob to null the zero-field Hall voltage; no offset correction was made in the analysis.

3.2 Procedure

Three measurements were made at room temperature, each repeated three times for each sample.

In the current sweep, the field was set to 250 mT and the control current was stepped from −30 mA to +30 mA in steps of about 5 mA, giving 13 points per run. One p-type run contains steps at −21 and −4 mA and a repeated point at 20 mA; all its points were used. The field was set before each run but was not recorded with the data.

In the field sweep, the control current was fixed at 30 mA and the field was stepped from −300 mT to +300 mT in steps of about 20 mT, with the coil current reversed at zero field. B, U_H and U_p were recorded at each of the 31 points per run.

In the temperature run, made once and only on the p-type sample, the field was set to 300 mT and the current to 30 mA, the heater was switched on, and U_H and the sample temperature were logged automatically as the sample warmed from 26 °C to 143 °C (59 points).

3.3 Analysis

All fits are unweighted linear least squares, with parameter uncertainties from the covariance scaled by the residual variance; the instrument resolutions (0.001 mV in U_H, 1 mV in U_p) are far below the observed scatter. The current sweeps were fitted with a straight line through all 39 (n-type) or 40 (p-type) points of the three runs. Each field sweep was fitted separately. The Hall voltage was fitted with a straight line over the full field range, as the manual prescribes. The resistance $R = U_p / I$ was fitted with $R = R_0 + \beta B + \gamma B^2$. The odd term βB absorbs the small part of the Hall voltage picked up by the sample-voltage contacts and changes μ_{MR} by less than 0.1%. Comparing with Eq. (4) gives $\mu_{\mathrm{MR}} = \sqrt{\gamma / R_0}$. For each run, H, the carrier density, σ_0 (Eq. (2)) and μ_H (Eq. (3), with $U_{p,0} = R_0 I$) were computed with the nominal L, w, d and I. Each quoted value is the mean of the three runs. Its uncertainty is the standard error of the three runs or, where larger, the propagated fit uncertainty divided by $\sqrt{3}$. Fitting the runs separately keeps the 1–2% drift of R_0 between runs (Sec. 4.3) out of the curvature.

The sensitivity of $\mathrm{d}U_H / \mathrm{d}B$ to the fit model was tested in two ways: with an odd cubic, $U_H = c_0 + s_0 B + s_3 B^3$, whose s_0 is the zero-field slope, and with a straight line restricted to $|B| \leq 100$ mT. The temperature of the sign reversal was found by linear interpolation between the two points bracketing $U_H = 0$.

The nominal dimensions and the set current carry uncertainties that the available records do not allow to be assessed. The carrier density scales as $1 / d$ and with I, and μ_H scales with L / w; the quoted uncertainties exclude these factors.

4. RESULTS

4.1 Carrier Sign

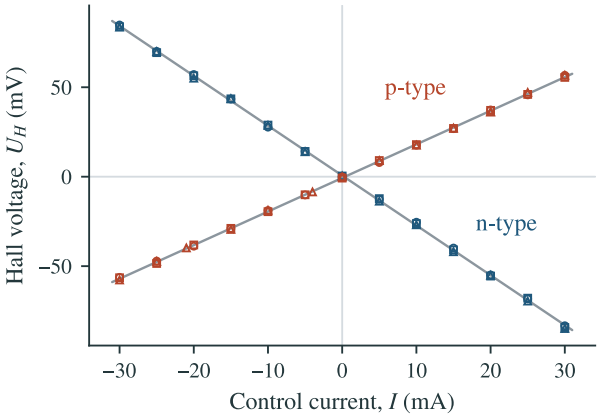


FIG. 3. Hall voltage against control current at a set field of 250 mT, for the n-type (blue) and p-type (red) samples. All points of the three runs are shown (circles, squares and triangles); grey lines are straight-line fits to all of them.

The Hall voltage was proportional to the control current, with opposite signs for the two samples (Fig. 3). The fitted slopes were (-2.787 ± 0.007) V/A for the n-type sample and $(+1.878 \pm 0.005)$ V/A for the p-type sample. The residual scatter about the lines was 0.8 mV and 0.5 mV respectively, about 1% of the largest Hall voltage. The n-type sample gave the negative slope that the manual shows for n-type Ge, where electrons carry the current [1], and the p-type sample the positive slope expected for holes.

4.2 Hall Coefficient, Carrier Density and Hall Mobility

The Hall voltage rose linearly with field (Fig. 4). The mean slopes were (-300 ± 2) mV/T for the n-type and (217.7 ± 0.9) mV/T for the p-type sample. At zero field the lines did not pass through the origin: the intercepts were (-9.73 ± 0.11) mV and $(+2.48 \pm 0.11)$ mV. Table I lists the quantities derived from the slopes and from the zero-field resistance.

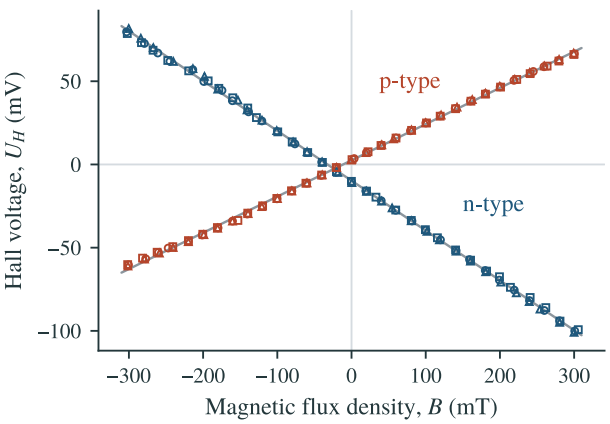


FIG. 4. Hall voltage against magnetic flux density at $I = 30$ mA, all points of the three runs (markers as in Fig. 3). Grey lines are straight lines with the mean slope and intercept of the three per-run fits. The lines do not pass through the origin; their intercepts are the zero-field offsets.

TABLE I. Results of the field sweeps at $I = 30$ mA: means of three runs with statistical uncertainties (Sec. 3.3). Mobilities are in $m^2\,V^{-1}\,s^{-1}$. The densities, H and σ_0 use the nominal $d = 1$ mm, and σ_0 and μ_H the nominal $L / w = 2$. The last row gives the change of dU_H / dB when it is taken as the zero-field slope s_0 of an odd cubic, and when it is fitted over $|B| \le 100$ mT only.

Quantity	n-type	p-type
dU_H / dB (mV T^{-1})	-300 ± 2	217.7 ± 0.9
H (10^{-3} m^3 C^{-1})	-9.99 ± 0.07	7.26 ± 0.03
Carrier density (10^{20} m^{-3})	6.25 ± 0.05	8.60 ± 0.03
R_0 (Ω)	55.1 ± 0.4	50.6 ± 0.2
σ_0 (S m^{-1})	36.3 ± 0.2	39.5 ± 0.2
μ_H	0.3628 ± 0.0015	0.2870 ± 0.0008
μ_{MR}	0.390 ± 0.010	0.565 ± 0.004
μ_{MR} / μ_H	1.07 ± 0.03	1.97 ± 0.01
Slope shift (%)	+1.0, −1.6	+5.5, +3.6

The straight line describes the n-type data to within the scatter, but the p-type residuals were systematic. In all three p-type runs the odd cubic gave $s_3 \approx -0.21$ V/T^3, so the Hall voltage of the p-type sample grew more slowly than linearly at high field. Its zero-field slope s_0 exceeded the full-range slope by 5.5%, and the slope over $|B| \le 100$ mT by 3.6%. The same tests moved the n-type slope by +1.0% and −1.6%, with s_3 of varying size in the three runs. For the p-type sample this model dependence, about five times the statistical uncertainty, dominates the uncertainty of H: the low-field values of p are 4–5% lower and of μ_H 4–6% higher than in Table I.

4.3 Magnetoresistance

The resistance of both samples rose quadratically with field, by (1.37 ± 0.07)% at 300 mT for the n-type and (2.87 ± 0.03) % for the p-type sample (Fig. 5). Within a run the quadratic described the data to 0.04–0.06 Ω (rms), close to the resolution of U_p. Between runs, however, the zero-field resistance changed by 2.2% (n-type) and 1.4% (p-type), comparable to the whole field-induced change. Averaging the runs before fitting would have mixed this drift into the curvature. The fitted mobilities were $\mu_{MR} = (0.390 \pm 0.010)$ and $(0.565 \pm 0.004)\mathrm{m^2V^{-1}s^{-1}}$, so that

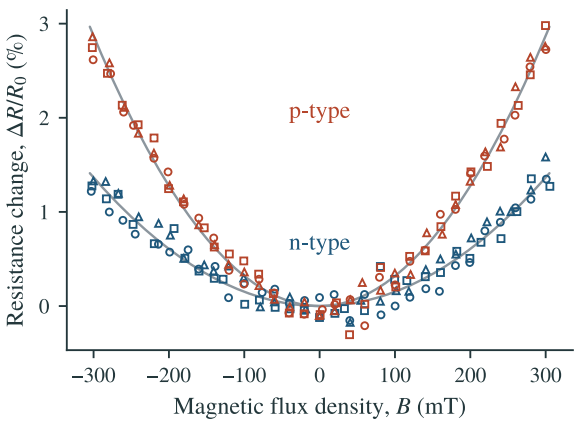


FIG. 5. Relative change of sample resistance with field at $I = 30$ mA, for each of the three runs (markers as in Fig. 3), with each run referred to its own fitted R_0 and its small odd term removed. The 1 mV resolution of U_p corresponds to 0.06%. Grey curves are $(\mu_{MR}B)^2$ with the mean fitted μ_{MR}.

$\mu_{MR} / \mu_H = 1.07 \pm 0.03$ for the n-type sample and 1.97 ± 0.01 for the p-type sample.

4.4 Temperature Dependence

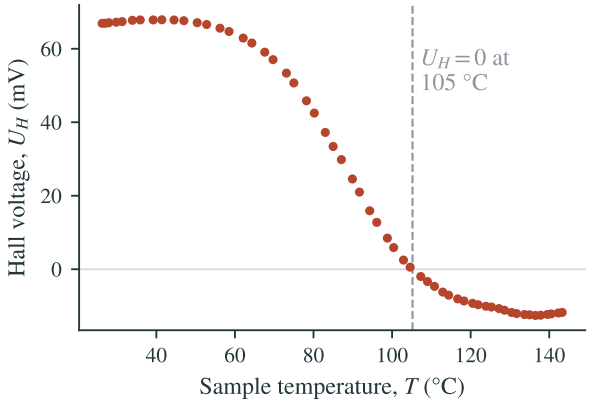


FIG. 6. Hall voltage of the p-type sample against sample temperature during heating, at $B = 300$ mT and $I = 30$ mA. The dashed line marks the interpolated sign reversal.

The Hall voltage of the p-type sample stayed near 67 mV up to about 45 °C, with a maximum of 67.9 mV at 41 °C, then fell steeply, changed sign at 105.2 °C, and reached −12.6 mV at 136 °C before rising slightly (Fig. 6). The reversal lies between points at 104.6 °C and 107.3 °C, where the Hall voltage fell by 1.0 mV/K. At the start of the run (26 °C) the Hall voltage was 66.9 mV, within 1.3% of the 67.8 mV predicted at 300 mT by the p-type field sweep.

5. DISCUSSION

5.1 Zero-Field Offset and the Current Sweeps

The current sweeps confirm the carrier signs but do not give an independent Hall coefficient, because their slopes are biased by the zero-field offset. Dividing the current-sweep slope by the field-sweep Hall slope per ampere implies a field during the current sweeps of (279 ± 2) mT for the n-type and (259 ± 1) mT for the p-type sample, both above the set 250 mT. The field-sweep intercepts show, however, a zero-field offset of –9.7 mV and +2.5 mV at 30 mA. Such an offset arises when the two Hall contacts are not exactly opposite each other, so that they also pick up part of the longitudinal voltage; it is then proportional to the current and independent of B. If an offset of the same size per ampere was present in the current sweeps, it adds −0.32 V/A and +0.08 V/A to their slopes. Subtracting these terms gives (247 ± 2) mT and (247.4 ± 1.3) mT. The two corrected fields agree with each other and lie within 1.5% of the set value, so the offset accounts for the whole discrepancy. The field sweeps, which record B at every point and separate the offset as an intercept, are therefore the basis of the Hall coefficients in Table I. The offset shows that the zero-field Hall voltage was not fully nulled.

5.2 Mobilities and Evidence for Two Hole Populations

The lattice-limited drift mobilities of Ge at 300 K are 0.39 $m^2\,V^{-1}\,s^{-1}$ for electrons and 0.19 $m^2\,V^{-1}\,s^{-1}$ for holes [5]. The

n-type Hall mobility is 7% below the electron value. The p-type Hall mobility is 51% above the hole value, and the ratio of the two Hall mobilities, 1.26, is well below the drift-mobility ratio of 2.05. Neither difference can come from the thickness, which cancels in Eq. (3), but both are subject to the nominal L / w and to the Hall factor r_H.

Three observations distinguish the p-type sample: its Hall mobility exceeds the drift value, its Hall voltage is sublinear in field, and its magnetoresistance mobility is twice its Hall mobility. The n-type sample shows none of these. A field-dependent Hall coefficient and a large magnetoresistance are both expected when carriers of very different mobility share the current [4], and the valence band of Ge contains light and heavy holes. The three p-type observations are therefore consistent with conduction by two hole populations. A high-mobility minority of light holes would raise the Hall mobility above the average drift mobility and make the Hall coefficient depend on field [4]. This experiment did not test that explanation directly. For the n-type sample, $\mu_{\mathrm{MR}} / \mu_H = 1.07 \pm 0.03$ shows that Eq. (4) gives roughly the right mobility in a single-carrier sample, without establishing why; the end-contact shorting and the scattering-time spread of Sec. 4.3 cannot be separated with one sample geometry.

5.3 Sign Reversal

The reversal temperature agrees with Eq. (7) to within the uncertainty of the inputs. Taking N_A as the measured room-temperature hole density, $8.6 \times 10^{20}\mathrm{m}^{-3}$, and the Ge parameters of Sze and Ng [5] in Eq. (6) ($n_i = 2.1 \times 10^{19}\mathrm{m}^{-3}$ at 300 K), Eq. (7) gives 99 °C for the 300 K drift-mobility ratio $b = 2.05$. It gives 117 °C and 91 °C for $b = 1.5$ and 2.5. The measured 105 °C lies within this range. The estimate cannot be sharper, because b itself changes with temperature and, at room temperature, the Hall mobilities of these samples differ in ratio from the drift values. Two effects bias the measured temperature. First, if the room-temperature offset of +2.5 mV persisted, the true zero of the Hall term lies where the measured voltage equals the offset, at most 2.5 K lower; since the offset scales with the resistance, which falls on heating, the actual shift is smaller. Second, the sensor and the sample may differ in temperature while heating, which this single run cannot assess.

5.4 Limitations

The densities rest on the nominal thickness and the set current, and the mobilities on the nominal length-to-width ratio. None of these was measured, so the absolute scale of the results is uncertain by an amount that could not be assessed; the comparison between the two samples, which share the board geometry, is less affected. Three runs give only a rough standard error. The 1–2% drift of R_0 between runs, of unknown origin, limits how well runs can be combined. For the p-type sample, the dependence of the slope on the fit model exceeds the statistical uncertainty.

6. CONCLUSION

The Hall voltages identified electrons as the majority carriers of the n-type sample and holes in the p-type sample. Field sweeps at 30 mA gave carrier densities of $(6.25 \pm 0.05) \times 10^{20}\mathrm{m}^{-3}$ and $(8.60 \pm 0.03) \times 10^{20}\mathrm{m}^{-3}$, and Hall mobilities of (0.3628 ± 0.0015) and $(0.2870 \pm 0.0008)\mathrm{m}^2\mathrm{V}^{-1}\mathrm{s}^{-1}$. The uncertainties are statistical; for the p-type sample, the fit range changes the result by a further 4–6%. The magnetoresistance mobilities were (0.390 ± 0.010) and $(0.565 \pm 0.004)\mathrm{m}^2\mathrm{V}^{-1}\mathrm{s}^{-1}$. They match the Hall mobility for the n-type sample but are twice it for the p-type sample. Together with the sublinear p-type Hall voltage, this is consistent with two hole populations of different mobility. The p-type Hall voltage changed sign at 105 °C, within the 91–117 °C range estimated from the measured hole density.

Four changes would address the limitations found here. Measuring the sample thickness, and the length and width between the voltage contacts, would turn the nominal scale of the densities and mobilities into a measured one with an uncertainty. Nulling the zero-field Hall voltage, or measuring at both field polarities and taking half the difference, would remove the offset, so that the current sweeps give an independent Hall coefficient. Recording the actual current, and the field during the current sweeps, would let the two methods be compared quantitatively. Extending the p-type field sweep to the full field the magnet can reach, with finer steps at low field, would test whether the Hall coefficient tends to a high-field limit, as two hole populations would predict.

7. REFERENCES

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